PHONEPE TRANSACTION INSIGHTS

Introduction:

This project analyzes PhonePe Pulse data to extract meaningful insights about transactions, user engagement, and market trends across India.

The dashboard is built using Streamlit and provides interactive visualizations.

Technologies Used:

- Python
- Pandas
- MySQL
- Streamlit
- Plotly
- Matplotlib & Seaborn

Data Source:

Data is taken from PhonePe Pulse GitHub repository and stored in MySQL database.

Database Structure:

Tables used:

-aggregated_transaction, aggregated_insurance, aggregated_user

-map_users, map_transaction, map_insurance

-top_transactions, top_insurance, top_users

Case Studies:

1. Decoding Transaction Dynamics on phonepe:
    - High-Value Transaction States:
      Identifies states with the highest average transaction amount, indicating regions with high-value digital payments.
    - Quarterly Growth Analysis:
      Analyzes transaction trends across quarters to understand seasonal patterns and growth fluctuations over time.
    - Transaction type cumulative trend:
      Displays how different transaction categories contribute cumulatively to overall transaction growth over the years.
    - Growth rate & decline detection:
      Tracks year-over-year growth percentage to identify periods of acceleration or decline in transactions.
    - High vs Low value transaction analysis:

      Classifies transactions into high and low value segments to compare their contribution to total transaction volume.

2. Transaction analysis for market expansion:
    - State-wise transaction growth analysis:
      Analyzes total transaction volume across states for a selected year to identify high-growth regions.
    - Low-performance state identification:
      Highlights states with below-average transaction activity to identify potential areas for improvement.
    - Year-wise market expansion trend:
      Tracks overall transaction growth over years to understand market expansion and adoption trends.
    - District-level penetration Analysis:
      Examines top-performing districts within a state to measure transaction penetration at a granular level.
    - Categorical-wise Transaction Analysis:
      Shows the distribution of transactions across different categories to understand user behavior and service usage patterns.

3. User Engagement and Growth Strategy:
    - High app opens-->Strong Engagement Regions:
      Identifies states with the highest app opens, indicating regions with strong user engagement and platform activity.
    - Category-wise user Engagement:
      Analyzes user activity distribution across different brands/categories to understand user preferences.
    - State-level Transaction inequality:
      Compares transaction volumes across states to highlight disparities between high-performing and low-performing regions.
    - Top Districts Driving Growth:
      Highlights top districts contributing the most to user growth based on registered users and app activity.
    - Quarterly App Engagement Trend by State:
      Tracks how user engagement (app opens) changes across quarters for each state to identify seasonal or growth patterns.
4. Insurance Engagement Analysis:
    - State-wise Insurance Adoption Analysis
    - User Engagement vs Transaction Trend
    - Low Engagement Region Identification
    - Quarterly growth trend analysis
    - Insurance product popularity analysis
5. Insurance Transactions Analysis:
    - Correlation Between Transaction count and amount
    - Insurance Penetration vs Population Density
    - Year-over-year Insurance Growth
    - Market Concentration Analysis
    - Average Insurance Transaction value by state

Conclusion:

This project provides a comprehensive understanding of digital payment trends and helps in making data-driven business decisions.